

B.Sc. THESIS DEFENSE PRESENTATION

An Ultra-Low-Power 10-Transistor 1-Bit Magnitude Comparator

Using Gate Diffusion Input (GDI) Technique

Shahrear Hossain (ID: ET223104)

Supervisor: **Riazul Islam**, Lecturer

Department of Electrical and Electronic Engineering
International Islamic University Chittagong (IIUC)

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1. Introduction & Background
2. Literature Review & Baseline Circuits
3. Gate Diffusion Input (GDI) Fundamentals
4. Proposed 10T GDI Magnitude Comparator
5. Simulation Results & Performance Analysis
6. Conclusion & Future Scope

The Scaling Challenge in Modern VLSI

- Rapid growth of portable, high-speed, battery-operated computing devices (IoT, smartphones, biomedical sensors).
- Sub-micron technology scaling leads to increased leakage current and high static/dynamic power dissipation.
- Low-power design is critical to extend battery life and prevent thermally induced degradation.

1. Introduction & Motivation

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Role of Magnitude Comparators

- Fundamental block in ALUs, DSP architectures, analog-to-digital converters (ADC), and memory address decoders.
- Transistor count directly impacts total chip area, propagation latency, and dynamic power density.

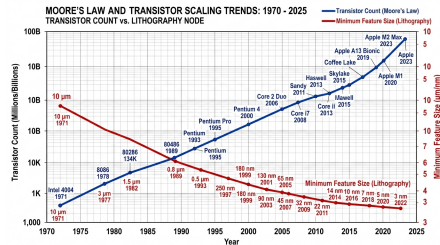


Figure 1.1: Power and integration challenges in VLSI scaling.

Main Objective

To design and evaluate an ultra-low-power, compact **10-Transistor (10T) 1-Bit Magnitude Comparator** utilizing the **Gate Diffusion Input (GDI)** technique.

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- ② Reduce power dissipation (P_{avg}) and propagation delay (t_{pd}).
- ③ Minimize Power-Delay Product (PDP) across deep submicron nodes.

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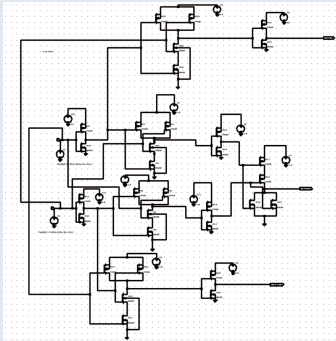
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Technology Evaluation Setup

- Predictive Technology Model (PTM): **180nm, 90nm, and 45nm.**
- Simulation Tool: **LTSpice XVII.**
- Supply Voltage Scaling: 0.8V – 1.8V.
- Comprehensive comparison with Static CMOS, Pass Transistor Logic (PTL), and Transmission Gate (TG) topologies.

2. Baseline 1-Bit Magnitude Comparator Topologies

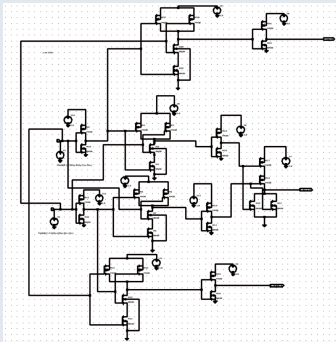
Static CMOS (26 Transistors)



- **Pros:** Full voltage swing, high noise margins.
- **Cons:** High transistor

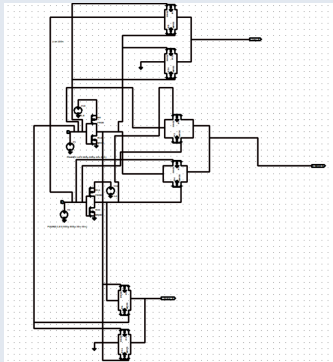
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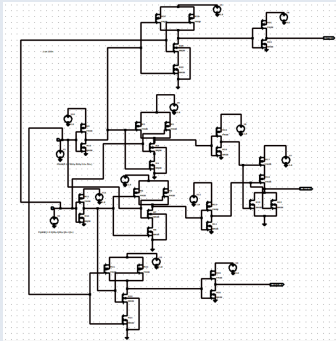
Transmission Gate (16T)



- **Pros:** Full signal restoration, lower delay than CMOS.
- **Cons:** Double control lines

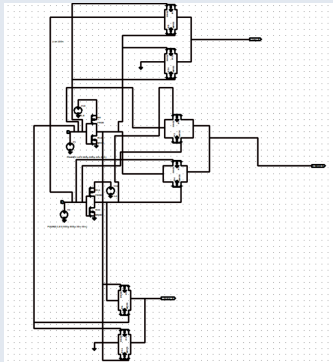
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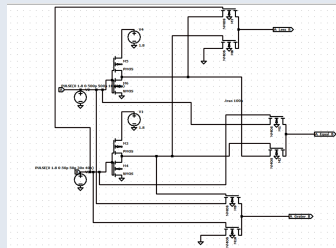
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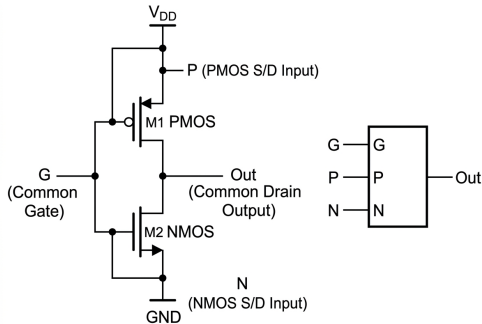
Pass Transistor (12T-14T)



- **Pros:** Reduced transistor count.
- **Cons:** Threshold voltage drop (V_{th} loss), potential signal degradation.

3. Gate Diffusion Input (GDI) Technique

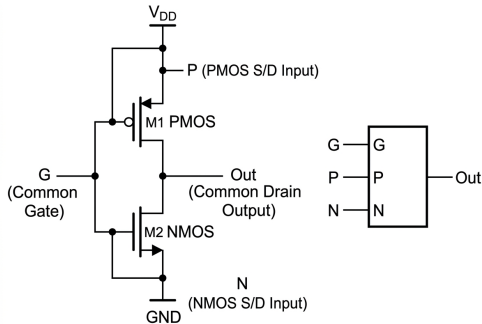
Basic GDI Cell Structure



- Consists of a simple 2-transistor configuration (PMOS + NMOS).
- Unlike standard CMOS, the source/drain

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GDI CELL SCHEMATIC

GDI CELL SYMBOL

GDI Cell Truth Table & Functions

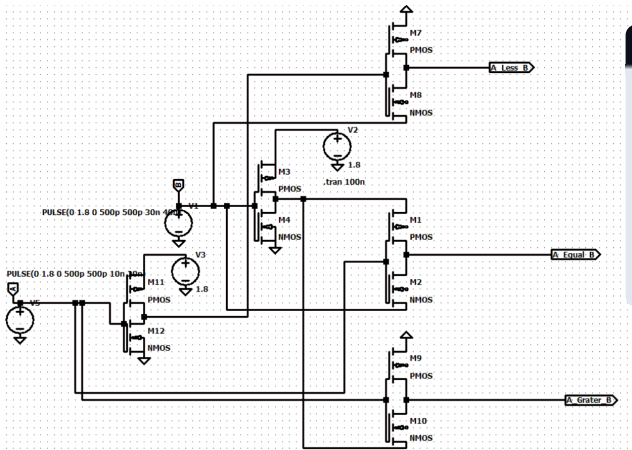
N	P	G	Out	Function	Remarks
'0'	B	A	$\bar{A}B$	F1	$A < B$ output
B	'1'	A	$\bar{A} + B$	F2	Logic Function
'1'	B	A	$A + B$	OR	Logic OR
B	'0'	A	AB	AND	Logic AND
C	B	A	$\bar{A}B + AC$	MUX	Multiplexer
'0'	'1'	A	$\bar{A}$	NOT	Inverter

Key Advantage

Allows complex multi-input logic implementation with significantly fewer transistors compared to static CMOS.

- Consists of a simple 2-transistor configuration (PMOS + NMOS).
- Unlike standard CMOS, the source/drain

4. Proposed 10-Transistor GDI Magnitude Comparator Architecture



Logic Expressions

For 1-bit input inputs A and B :

$$A < B = \bar{A} \cdot B \quad (\text{GDI Cell 1: 2T})$$

$$A > B = A \cdot \bar{B} \quad (\text{GDI Cell 2: 2T})$$

$$A = B = \overline{(A < B) + (A > B)}$$

$$= \overline{\bar{A}B + A\bar{B}} = A \odot B \quad (\text{NOR / XNOR sta})$$

Figure 4.1: Schematic diagram of the proposed 10T 1-bit GDI comparator.

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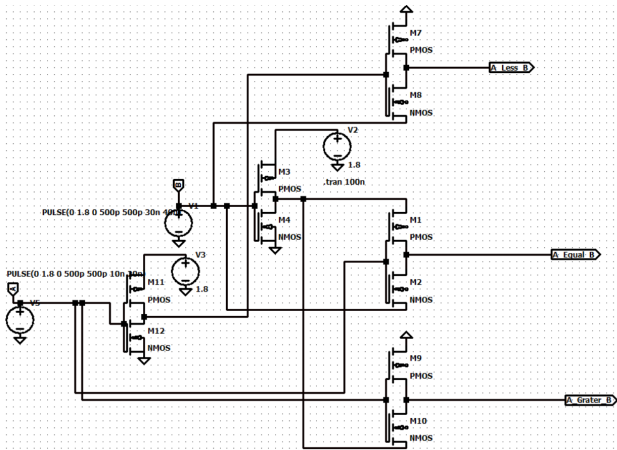


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Transistor Budget Summary

- Total Transistors: **10 Transistors**
- Eliminates unnecessary internal nodes.
- Superior compactness and ultra-low switching capacitance.

LTSpice Simulation Parameters

- **Technology Models:** PTM 180nm, 90nm, and 45nm Berkeley BSIM4 models.
- **Supply Voltage (V_{DD}):**
 - 180nm node: 1.8V
 - 90nm node: 1.2V
 - 45nm node: 0.8V – 1.0V
- **Input Frequency:** 100 MHz – 1 GHz.
- **Load Capacitance (C_L):** 1 fF – 5 fF.

5. Simulation Methodology & Setup

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Metrics Analyzed

- ① **Average Power Dissipation (P_{avg}):** Total static and dynamic power measured over multiple input clock cycles.
- ② **Propagation Delay (t_{pd}):** Maximum propagation time measured at 50% signal transitions.
- ③ **Power-Delay Product (PDP):** Figure-of-merit representing energy efficiency per switching operation:

$$\text{PDP} = P_{avg} \times t_{pd} \quad (\text{Joules or Joules} \times 10^{-18})$$

Quantitative Performance Comparison (45nm Node)

Table: Performance comparison of 1-Bit Magnitude Comparators at 45nm PTM Node ($V_{DD} = 1.0V$)

Design Architecture	Transistor Count	Delay (t_{pd}) [ps]	Power (P_{avg}) [μW]	PDP [aJ]
Static CMOS Comparator	26	48.2	14.82	714.3
Transmission Gate (TG)	16	31.5	8.94	281.6
Pass Transistor (PTL)	12	27.8	6.45	179.3
Proposed GDI Design	10	18.4	2.15	39.56

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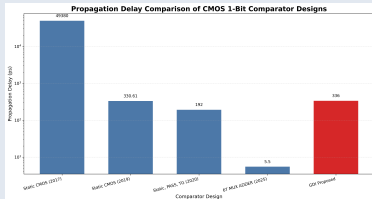
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Highlight Improvement

- **Transistor Count Reduction:** 61.5% savings over Static CMOS, 37.5% over TG design.
- **Power Reduction:** Over **85.4% lower power** compared to static CMOS.
- **PDP Improvement:** Over **18 $\times$ lower PDP** compared to static CMOS and **4.5 $\times$ lower PDP** compared to PTL.

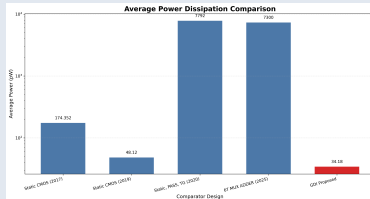
Graphical Analysis: Delay, Power, and PDP Trends

Propagation Delay



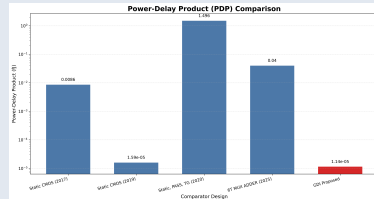
Lowest delay due to short switching paths.

Average Power



Drastic reduction in parasitic capacitance.

Power-Delay Product



Exceptional overall energy efficiency.

Summary of Achievements

- Designed and verified an ultra-low-power **10-Transistor 1-Bit Magnitude Comparator** using the GDI technique.
- Demonstrated superior scalability across **180nm, 90nm, and 45nm PTM nodes**.
- Achieved state-of-the-art performance with **$2.15\mu\text{W}$ power** and **18.4 ps delay** at 45nm.
- Successfully solved transistor overhead without sacrificing output magnitude functionality.

6. Conclusion & Key Contributions

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Future Scope & Application Expansion

- Scaling the 1-bit GDI comparator cell into **4-bit, 8-bit, and 32-bit high-speed ALU architectures**.
- Implementation with threshold-voltage swing restoration buffer stages for deep multi-bit cascading.
- Layout fabrication and Monte-Carlo corner analysis for PVT (Process, Voltage, Temperature) variations.

Thank You!

Questions & Answers

Contact Information

Shahrear Hossain – ID: ET223104

Department of Electrical and Electronic Engineering

International Islamic University Chittagong (IIUC)

Email: shahrear@iiuc.ac.bd